

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15E.1

SIMPLE CHROMATOGRAPHY (PAPER & TLC)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E.1: Simple Chromatography (Paper & TLC)
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15E.1

Q1. 1. (a) Define R_f value in thin-layer chromatography (TLC) and state its mathematical formula.

[2 Marks]

WCH14/01/Jan22/Q5(a)

Q2. (b) Explain why different amino acids have different R_f values on a silica TLC plate developed in a polar solvent.

[3 Marks]

WCH14/01/Jan22/Q5(b)

SECTION B: WORKED MARK SCHEME — 15E.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Define R _f value in thin-layer chromatography (TLC) and state its mathematical formula. 1. $R_f = (\text{distance travelled by spot}) / (\text{distance travelled by solvent front})$. 2. Ratio of solute movement relative to solvent.	[2]
Q2	Q2: (b) Explain why different amino acids have different R _f values on a silica TLC plate developed in a polar solvent. 1. Stationary phase (silica) is polar; mobile phase (solvent) is polar/non-polar. 2. Amino acids adsorb onto stationary phase and dissolve in mobile phase. 3. Amino acid with greater affinity/solubility in mobile phase moves faster → larger R _f value.	[3]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15E.1**FAQ #1: Why must the solvent level be below the baseline spot in TLC?****Category: Experimental Technique****COMMON EXAMINER TRAP / MISTAKE:**

Stating the solvent level doesn't matter.

MODEL ANSWER / EXAMINER EXPECTATION:

If the solvent level is above the baseline, the sample spots will dissolve directly into the bulk solvent reservoir rather than moving up the plate with the mobile phase.